

Autonomous Traffic Sign Recognition Using Custom CNN and Transfer Learning with ResNet18 on GTSRB

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Abstract

This paper presents a deep learning pipeline for autonomous traffic sign recognition on the German Traffic Sign Recognition Benchmark (GTSRB), a 43-class image classification benchmark comprising approximately 50,000 real-world photographs. Two architectures are designed, trained, and compared: a custom four-block Convolutional Neural Network (CNN) built from scratch, and a ResNet18 model adapted via partial fine-tuning of its final residual block. Both models are evaluated on the held-out test set using accuracy, per-class precision, recall, F1-score, and confusion matrices. The custom CNN achieves 81.3% test accuracy after 20 training epochs with aggressive regularization, while the partially fine-tuned ResNet18 reaches 94.2% test accuracy in only 8 epochs, with a macro-averaged F1-score of 0.92. A perception demonstration is also provided, showing the classification pipeline applied to individual test-set images with bounding box overlays.

1 Introduction

Traffic sign recognition is a critical perception module in autonomous driving systems. A reliable classifier must handle 43 semantically distinct classes that share visual characteristics — circular red borders, triangular warning frames, and pictograms that differ only in minor details (e.g., *Speed Limit 30 km/h* vs. *Speed Limit 50 km/h*). Misclassification of safety-critical classes such as STOP or No Entry can have severe real-world consequences, setting strict requirements on both accuracy and robustness.

The German Traffic Sign Recognition Benchmark (GTSRB) [1] is the standard public dataset for this task. It contains over 50,000 images captured under varying lighting conditions, camera distances, and partial occlusions, making it a representative proxy for deployment conditions.

This work addresses two complementary research questions: (1) Can a custom CNN trained from scratch, with appropriate regularization, achieve competitive accuracy on a 43-class fine-grained benchmark? (2) Does partial fine-tuning of a pretrained ResNet18 outperform full feature extraction on this domain, and if so, by how much?

2 Related Work

Convolutional neural networks were applied to GTSRB as early as 2012, when Ciresan et al. [2] achieved superhuman performance using a multi-column deep CNN. Since then, the benchmark has served as a standard testbed for data augmentation strategies [5], transfer learning pipelines [4], and lightweight architectures for embedded deployment. ResNet architectures [3] introduced residual connections that address the vanishing gradient problem and enable stable training of deeper networks. Transfer learning from ImageNet [6] has become the dominant paradigm for natural image classification, but its effectiveness on domain-specific datasets — where target classes differ substantially from ImageNet categories — depends heavily on the degree of backbone fine-tuning.

3 Methodology

3.1 Dataset and Preprocessing

GTSRB images are distributed as .ppm files organized into 43 class folders for training and a flat directory for testing. The pipeline downloads three archives from the official ERDA mirror, converts all images to JPEG, and constructs three non-overlapping splits: 90% of each class for training, 10% for validation, and the provided test partition (12,630 images) for final evaluation. The test set is used exclusively at evaluation time — never during training or hyperparameter selection.

All images are resized to 224×224 pixels and normalized with ImageNet statistics (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]). Two separate augmentation pipelines are applied during training:

- **CNN pipeline:** `RandomRotation(15°)`, `RandomAffine(translate  $\pm 10\%$ , scale 0.9–1.1)`, `ColorJitter(brightness/contrast 0.4, saturation 0.3)`.
- **ResNet18 pipeline:** lighter augmentation (`RandomRotation(10°)`, `ColorJitter(0.2)`) to avoid distorting ImageNet features.

Horizontal flipping is deliberately excluded because

mirroring changes the semantic meaning of directional signs (e.g., Turn Left $\rightarrow$ Turn Right). Batch size is 64 for both models.

3.2 Model A: Custom CNN from Scratch

The custom architecture follows a VGG-style design with four convolutional blocks. Each block applies $\text{Conv}3\times3 \rightarrow \text{BatchNorm} \rightarrow \text{ReLU} \rightarrow \text{MaxPool}2\times2$, doubling the channel depth at each stage:

$$3 \xrightarrow{B1} 32 \xrightarrow{B2} 64 \xrightarrow{B3} 128 \xrightarrow{B4} 256$$

An `AdaptiveAvgPool2d(1)` layer collapses spatial dimensions to a 256-dimensional feature vector, which is passed through a two-layer classifier ($\text{Linear}(256,128) \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.4) \rightarrow \text{Linear}(128,43)$). `BatchNorm` after every convolution stabilizes training and reduces sensitivity to the learning rate. The model has 427,819 trainable parameters.

To mitigate overfitting, L2 regularization is applied via `weight_decay=1e-3` in the Adam optimizer. Training runs for 20 epochs with `CosineAnnealingLR` ($T_{\max} = 20$, $\text{lr} = 10^{-3}$).

3.3 Model B: ResNet18 with Partial Fine-Tuning

A ResNet18 pretrained on ImageNet is adapted for GT-SRB using a *partial fine-tuning* strategy. The first three residual blocks (`layer1-layer3`) remain frozen, preserving low-level ImageNet features (edges, textures) that generalize across natural image domains. The final block (`layer4`) is unfrozen with learning rate 10^{-4} , allowing the model to adapt high-level representations to traffic sign geometry. The classification head is replaced with a `Dropout(0.3) + Linear(512, 43)` module, trained at learning rate 10^{-3} .

This strategy was preferred over pure feature extraction (fully frozen backbone) because GT-SRB has 43 visually similar classes that share colors and shapes — frozen ImageNet features proved insufficient to discriminate them (preliminary experiments yielded only $\sim 63\%$ validation accuracy with a frozen backbone vs. $\sim 90\%$ from the first epoch with `layer4` unfrozen). Approximately 72% of total parameters are trainable under this configuration (8,415,787 out of 11.2M). Training runs for 8 epochs with the same `CosineAnnealingLR` scheduler.

4 Experiments

4.1 Training Dynamics

The custom CNN shows a steady learning curve across 20 epochs, with validation accuracy tracking training accuracy closely — evidence of effective regularization.

Minor oscillations in validation loss (epochs 13–15) correspond to the cosine annealing schedule and do not indicate pathological overfitting, as validation loss resumes its downward trend at epoch 16.

ResNet18 converges dramatically faster: it reaches 90% validation accuracy in the very first epoch due to the pre-learned feature representations. The generalization gap (train 99% vs. val 95%) remains stable across all 8 epochs, confirming that `Dropout(0.3)` and `weight_decay=1e-4` effectively limit overfitting despite the high number of trainable parameters.

4.2 Test Set Results

Table 1 summarizes quantitative performance on the held-out test partition.

Table 1: Test set performance comparison.

Metric	Custom CNN	ResNet18
Test Accuracy	81.3%	94.2%
Macro Avg F1	0.73	0.92
Weighted Avg F1	0.81	0.94
Trainable Parameters	427,819	8,415,787
Training Epochs	20	8

The 12.9 percentage-point accuracy gap reflects the advantage of pretrained features for a fine-grained 43-class problem. The small difference between macro and weighted F1 for ResNet18 (0.92 vs. 0.94) indicates consistent performance across both frequent and rare classes, with no systematic bias toward majority classes.

4.3 Perception Demonstration

Section 11 of the notebook demonstrates the classification pipeline applied to individual images randomly sampled from the test set. Each image is classified by ResNet18, and the predicted label and confidence score are overlaid as a bounding box. Box color encodes confidence: green ($>85\%$), yellow (50-85%), red ($<50\%$). This demonstrates the complete output format expected in an autonomous vehicle perception system.

5 Conclusions

This paper demonstrates two complementary approaches to traffic sign recognition on GT-SRB. The custom CNN, built from scratch with four convolutional blocks and aggressive regularization, achieves 81.3% test accuracy — a competitive result given its compact size (427K parameters), making it suitable for edge deployment on embedded hardware. The partially fine-tuned ResNet18 achieves 94.2% test accuracy in fewer than half the training epochs, confirming that partial fine-tuning is significantly more effective than pure feature extraction for

fine-grained multi-class benchmarks where frozen ImageNet features are insufficient.

Key design decisions with measurable impact include: (i) separating augmentation pipelines for CNN and ResNet18 (heavier distortions for the randomly initialized model, lighter for the pretrained backbone); (ii) unfreezing only the last ResNet block, which improved first-epoch validation accuracy from 55% to 90%; and (iii) using `CosineAnnealingLR` to stabilize convergence in final epochs.

Future work could explore knowledge distillation from ResNet18 to a compact student network, or replace the classification-only pipeline with a dedicated detector (e.g., YOLOv8) for full end-to-end localization and recognition in unconstrained road scenes.

References

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